

SUPERSTORE SALES & PROFIT ANALYSIS

Retail Business Performance Analysis | MySQL + Power BI

Project Overview

This project analyzes transactional retail sales data to identify revenue and profit drivers, compare category and regional performance, evaluate discount impact, and highlight profitable and loss-making products and customers. The analysis was performed with SQL in MySQL and presented through an interactive Power BI dashboard.

Tools & Deliverables

Tools: MySQL, SQL, Power BI

Deliverables: Power BI dashboard, SQL analysis file, dashboard walkthrough video, and this PDF report.

Key Performance Indicators

Metric	Result
Total rows analyzed	9,694
Total sales	\$2,272,449.86
Total quantity	36,749
Total profit	\$282,857.75
Average discount	15.52%
Overall profit margin	12.45%

Category Performance

Technology generated the highest sales and profit among the three categories, with approximately \$835.9K in sales and \$145.4K in profit. **Office Supplies** produced approximately \$703.5K in sales and \$120.5K in profit. **Furniture** generated approximately \$733.0K in sales but only \$17.0K in profit, showing a much weaker margin than the other categories.

Regional Performance

The **West** region led regional profitability with approximately \$106.0K in profit and \$713.5K in sales. The **East** followed with approximately \$90.7K profit. The **Central** region had the lowest profit margin at about 8.06%, indicating an area where pricing, discounting, product mix, or operating efficiency should be reviewed.

Discount & Profitability Analysis

The analysis shows that higher discounts can materially reduce profitability for selected products. The Tables sub-category is a clear example: at 0% discount it produced positive profit, while higher discount levels produced increasingly negative profit. This indicates that discount policy should be evaluated by product and margin rather than applied uniformly.

Product & Customer Analysis

The dashboard identifies the most profitable products and the top loss-making products, allowing management to focus on high-value products while reviewing products that consistently destroy margin. Customer-level analysis also highlights both high-profit customers and customers associated with significant losses.

Dashboard Structure

Page 1 — Executive Summary: KPI cards for sales, profit, quantity and profit margin, plus category, regional, monthly, and top-product analysis.

Page 2 — Detailed Analysis: discount vs. profit, ship mode, category, sub-category, customer segment, regional profit, and category-level profit margin analysis.

Business Recommendations

1. **Review Furniture profitability:** investigate low-margin products, pricing, freight, and discount levels within Furniture.
2. **Control deep discounting:** set product-level discount thresholds, especially where discounts consistently convert positive margins into losses.
3. **Focus on profitable products:** protect availability and visibility of high-profit products while reviewing persistently loss-making items.
4. **Investigate Central region performance:** identify whether product mix, pricing, discounts, or other operational factors explain its lower margin.
5. **Use customer profitability:** prioritize high-value relationships and investigate customers with repeated negative profitability.

Project Conclusion

The analysis demonstrates how SQL-based retail data analysis can be transformed into an actionable Power BI dashboard. The project provides a clear view of where sales are generated, where profit is created or lost, and which product, customer, category, region, and discount patterns deserve management attention.

Scope note: The completed project uses MySQL and Power BI. The original concept mentioned Python/Seaborn, Tableau, inventory turnover, and seasonal analysis; those components were not included in the completed implementation and are therefore not represented as completed work in this report.